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Q1 A1

4 points

Q1.1 (a)(i)

0.5 points

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i. $\lim_{t \rightarrow 0^-} g(t) =$

⋮ 1 -0.0

Correct. The answer is -1 .

⋮ 2 -0.5

Incorrect. The answer is -1 .

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Q1.2 (a)(ii)

0.5 points

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ii. $\lim_{t \rightarrow 0} g(t) =$

⋮ -0.0

Correct. The answer is DNE.

⋮ -0.5

Incorrect. The answer is DNE.

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Q1.3 (a)(iii)

0.5 points

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iii. $\lim_{t \rightarrow 2^+} g(t) =$

⋮ -0.0

Correct. The answer is 0.

⋮ -0.5

Incorrect. The answer is 0.

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Q1.4 (a)(iv)

0.5 points

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iv. $\lim_{t \rightarrow 4} g(t) =$



1

-0.0

Correct. The answer is 3.



2

-0.5

Incorrect. The answer is 3.

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Q1.5 (b)(i)

0.5 points

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i. $\lim_{x \rightarrow 0} \frac{\sin^2(x)}{x \tan(x)}$



1

-0.0

Correct. The answer is 1.



2

-0.5

Incorrect, or insufficient or incorrect reasoning.

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Q1.6 (b)(ii)

0.5 points

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ii. $\lim_{x \rightarrow -\infty} \frac{\sqrt{-x^3 + x^2 - x + 1}}{10x - 3}$

⋮ 1 -0.0

Correct. The answer is $-\infty$.

⋮ 2 -0.5

Incorrect, or insufficient or incorrect reasoning. The answer is $-\infty$.

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Q1.7 (c)

1 point

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(c) Find the value of a for which the function $f(x)$ given on the right is continuous at $x = 1$.

$$f(x) = \begin{cases} x^2 - a & \text{if } x \leq 1 \\ \frac{x^2 - 1}{x^2 + 2x - 3} & \text{if } x > 1 \end{cases}$$

⋮ 1 -0.0

Correct. The answer is $a = \frac{1}{2}$.

❖ 2 -0.5

Some meaningful progress is made toward a solution, but there are still significant errors.

❖ 3 -1.0

No meaningful progress is made toward a solution.

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Q2 A2

4 points

Q2.1 (a)

2 points

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(a) Use the limit definition of the derivative to show that if $f(x) = mx + b$, then $f'(x) = m$.

❖ 1 +2.0

Correct

**2****+0.5**

Some meaningful progress is made toward setting up the limit, but there are still significant mistakes.

**3****+1.0**

The appropriate limit is setup.

**4****+0.5**

Some meaningful progress is made toward evaluating the limit, but there are still significant mistakes. The correct answer shows that

$$\lim_{h \rightarrow 0} \frac{m(x+h) + b - (mx+b)}{h} = m.$$

**5****+1.0**

The limit is correctly evaluated. The correct answer shows

that $\lim_{h \rightarrow 0} \frac{m(x+h) + b - (mx+b)}{h} = m.$

**6****+0.0**

Little to no progress is made toward a solution or no work is submitted.

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Q2.2 (b)

2 points

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(b) Let $g(x) = \frac{k}{x^2}$ for some constant k . Use the limit definition of the derivative to find $g'(x)$.



1

+2.0

Correct



2

+0.5

Some meaningful progress is made toward setting up the limit, but there are still significant mistakes.



3

+1.0

The correct limit is setup.



4

+0.5

Some meaningful progress is made toward evaluating the limit, but there are still significant errors. The correct

answer shows that $\lim_{h \rightarrow 0} \frac{\frac{k}{(x+h)^2} - \frac{k}{x^2}}{h} = \frac{-2k}{x^3}$.



5

+1.0

The limit is correctly evaluated. The correct answer shows

that $\lim_{h \rightarrow 0} \frac{\frac{k}{(x+h)^2} - \frac{k}{x^2}}{h} = \frac{-2k}{x^3}.$

⋮ +0.0

No meaningful progress or no work is submitted.

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Q3 A3

4 points

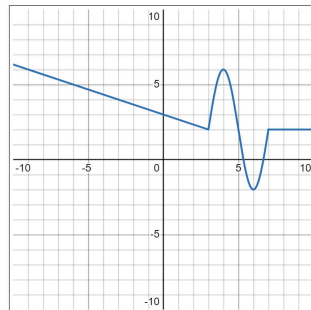
Q3.1 (a)

2 points

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Consider the graph of a function $f(x)$ shown to the right. Use this graph to answer each of the following questions.

- (a) Let $g(x)$ be the function $g(x) = \frac{f(\sin(-3x))}{-2e^{-x}}$. Find $g'(0)$.



⋮ +2.0

Correct

⋮ +0.5

Some meaningful progress is made toward finding the correct derivative, but there are still significant mistakes.

The correct answer is

$$\frac{6e^{-x}f'(\sin(-3x))\cos(-3x) - 2e^{-x}f(\sin(-3x))}{4e^{-2x}}.$$

3 +1.0

The correct derivative is found. The correct answer is

$$\frac{6e^{-x}f'(\sin(-3x))\cos(-3x) - 2e^{-x}f(\sin(-3x))}{4e^{-2x}}.$$

4 +0.5

Some meaningful progress is made toward evaluating the derivative at $x = 0$, but there are still significant mistakes (e.g. correctly found $f'(0) = -\frac{1}{3}$). The correct answer is -2 .

5 +1.0

The derivative is correctly evaluated at $x = 0$. The correct answer is -2 .

6 +0.0

No meaningful progress or no work submitted.

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Q3.2 (b)

2 points

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(b) Let $h(x) = x^{f(x)}$. Find $h'(e^2)$. It may be helpful to know that $e^2 \approx 7.4$

⋮ 1 +2.0

Correct

⋮ 2 +0.5

Some meaningful progress is made toward finding the correct derivative, but there are still significant mistakes.

The correct answer is $e^{f(x) \ln(x)} \left(f'(x) \ln(x) + \frac{f(x)}{x} \right)$.

⋮ 3 +1.0

The derivative is found correctly. The correct answer is

$e^{f(x) \ln(x)} \left(f'(x) \ln(x) + \frac{f(x)}{x} \right)$.

⋮ 4 +0.5

Some meaningful progress is made toward evaluating the derivative at $x = e^2$, but there are still significant mistakes. The correct answer is $2e^2$.

⋮ 5 +1.0

The derivative is evaluated correctly at $x = e^2$. The correct answer is $2e^2$.

⋮ 6 +0.0

No meaningful progress or no work submitted.

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Q4 A4

4 points

Q4.1 (a)

2 points

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(a) Consider the implicit curve given by the equation

$$x^2 + xy + y^2 = 13$$

This curve defines an implicit function $f(x)$ such that $f(4) = -1$ and another implicit function $g(x)$ such that $g(4) = -3$. Find both $f'(4)$ and $g'(4)$.

⋮ 1 +2.0

Correct. The correct answer is $f'(4) = -\frac{7}{2}$ and $g'(4) = \frac{5}{2}$.

⋮ 2 +0.5

Some meaningful progress is made toward completing the implicit differentiation, but there are still significant

mistakes.

⋮ 3 +1.0

The implicit differentiation is done correctly:

$$\frac{dy}{dx} = \frac{-y - 2x}{x + 2y}$$

⋮ 4 +0.5

Some meaningful progress is made toward evaluating the derivatives at $(4, -1)$ and $(4, -3)$, but there are still significant mistakes.

⋮ 5 +1.0

The values $(x, y) = (4, -3)$ and $(x, y) = (4, -1)$ are correctly substituted to find $g'(4) = \frac{5}{2}$ and $f'(4) = -\frac{7}{2}$.

⋮ 6 +0.0

Little to no progress is made toward a solution.

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2 points

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- (b) The number $\frac{4}{(1.9)^2}$ can be approximated using the linearization of the function $f(x) = \frac{4}{x^2}$ at $x = 2$. Find this approximation.

⋮ 1 +2.0

Correct. The answer uses the linear approximation at $a = 2$ to find $f(1.9) \approx 1.1$.

⋮ 2 +0.5

The answer finds $f'(2) = -1$.

⋮ 3 +0.5

The answer finds $f(2) = 1$.

⋮ 4 +0.5

The answer finds $L(x) = f'(2)(x - 2) + f(2)$. Credit should be awarded if this step follows correctly from an incorrect previous step.

⋮ 5 +0.5

The answer evaluates $L(1.9)$. Credit should be awarded if this step follows correctly from an incorrect previous step.



6

+0.0

Little to no progress is made toward a solution.

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Q5 A5

4 points

Q5.1 (a)

2 points

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(a) At what rate is the volume of a cylinder changing if its radius is 1 cm and decreasing at a rate of 2 cm per second, and its height is 3 cm and increasing at a rate of 12 cm per second? The volume of a cylinder of radius r and height h is $V = \pi r^2 h$.



1

+2.0

Correct



2

+0.5

Some meaningful progress is made toward the time derivative of V , but there are still significant mistakes.

The correct answer is $\frac{dV}{dt} = \pi \left(r^2 \frac{dh}{dt} + 2r \frac{dr}{dt} h \right)$.



3

+1.0

The time derivative of V is found correctly. The correct answer is $\frac{dV}{dt} = \pi \left(r^2 \frac{dh}{dt} + 2r \frac{dr}{dt} h \right)$.

⋮ 4 +0.5

The given values are used along with the time derivative equation to evaluate $\frac{dV}{dt} = 0$.

⋮ 5 +0.5

The correct units are reported.

⋮ 6 +0.0

No meaningful progress or no work is submitted.

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Q5.2 (b)

2 points

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(b) If two resistors with resistances R_1 and R_2 are connected in parallel, then the total resistance R , measured in ohms (Ω) is given by

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

If R_1 and R_2 are increasing at rates of $7 \Omega/s$ and $5 \Omega/s$ respectively, how fast is R changing, in Ω/s , when $R_1 = 2 \Omega$ and $R_2 = 1 \Omega$

⋮ 1 +2.0

Correct

⋮ 2 +0.5

Some meaningful progress is made toward the time derivative of V , but there are still significant mistakes.

The correct answer is

$$-R^{-2} \frac{dR}{dt} = -R_1^{-2} \frac{dR_1}{dt} - R_2^{-2} \frac{dR_2}{dt}.$$

⋮ 3 +1.0

The time derivative of V is found correctly. The correct

answer is $-R^{-2} \frac{dR}{dt} = -R_1^{-2} \frac{dR_1}{dt} - R_2^{-2} \frac{dR_2}{dt}.$

⋮ 4 +0.5

The given values are used along with the time derivative equation to evaluate $\frac{dR}{dt} = 3.$

⋮ 5 +0.5

The correct units are reported.

⋮ 6 +0.0

No meaningful progress or no work is submitted.

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